

# Chapter 5: Expected Utility Theory

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ECON 5001

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Based on: *Game Theory* by Giacomo Bonanno

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# Beyond Risk Neutrality

Chapter 4: chance moves give us **lotteries** over outcomes

Our shortcut was to assume:

- All outcomes are money
- Players are **risk neutral** (rank lotteries by expected value)

That leaves out a lot:

- People buy insurance. (And lottery tickets!)
- Many outcomes aren't money at all
  - Win/lose/draw, getting the job, where to eat dinner...

Our goal:

1. Rank lotteries over **any** kind of outcome
2. Describe different attitudes toward risk

## MobLab: “Bomb Risk Game”

- MobLab time!
- There are 100 boxes. One of them hides a **bomb**
  - Every box is equally likely to have it
- You choose how many boxes to open (anything from 0 to 100)
- Then we find out where the bomb was:
  - **Not** in your boxes: you earn 1 point per box you opened
  - **In** your boxes: you earn 0

Feel free to talk! And play it the way you'd actually play it.

## How Many Boxes?

Fewest boxes opened: \_\_\_\_\_

Median # of boxes opened: \_\_\_\_\_

Most boxes opened: \_\_\_\_\_

# Lotteries

# Lotteries

A **lottery**  $L$  is a list of outcomes and their probabilities:

$$L = \begin{pmatrix} o_1 & o_2 & \cdots & o_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}$$

The outcomes can be **anything**:

- Money:  $L_1 = \begin{pmatrix} \$10 & \$20 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

- Not money:  $L_2 = \begin{pmatrix} \text{Buckeyes beat Michigan} & \text{Buckeyes lose to Michigan} \\ \frac{3}{5} & \frac{2}{5} \end{pmatrix}$

A sure thing is a lottery too. Just write  $o$  for “outcome  $o$  with probability 1”

## Expected Value: Numbers Only

Recall the **expected value** of a money lottery:

$$E[L] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

$$L_1 = \begin{pmatrix} \$10 & \$20 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \quad E[L_1] = \underline{\hspace{2cm}}$$

$$L_2 = \begin{pmatrix} \text{Beat Michigan} & \text{Lose to Michigan} \\ \frac{3}{5} & \frac{2}{5} \end{pmatrix} \quad E[L_2] =$$

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**Lesson:** We can calculate an expected value when outcomes are **numbers** (like money).  
When outcomes are **not** numbers, expected value doesn't exist!

So how do we rank lotteries like  $L_2$ ?

## **Expected Utility**

# Two Kinds of Utility

**Step 1:** Give every *outcome* a number

- $u(o)$  is the **utility index** of outcome  $o$  (small  $u$ : outcomes)

**Step 2:** Use those to give every *lottery* a number

- $U(L)$  is the **expected utility** of lottery  $L$  (capital  $U$ : lotteries)

Expected utility: average the utility index, weighting by the probabilities.

$$\underbrace{U(L)}_{\text{expected utility of the lottery}} = \underbrace{p_1 u(o_1) + p_2 u(o_2) + \cdots + p_n u(o_n)}_{\text{probability-weighted average of the utility index}}$$

Bonanno writes  $U(o)$  for our  $u(o)$ , and  $E[U(L)]$  for our  $U(L)$ .

## Example

Alex's utility index over dinner outcomes:

| Outcome $o$          | Cazuela's | Yoshi Sushi | Dining hall |
|----------------------|-----------|-------------|-------------|
| Utility index $u(o)$ | 10        | 6           | 0           |

Two lotteries:

$$L_A = \begin{pmatrix} \text{Yoshi Sushi} \\ 1 \end{pmatrix} \qquad L_B = \begin{pmatrix} \text{Cazuela's} & \text{Dining hall} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$U(L_A) =$$

$$U(L_B) =$$

Alex prefers: \_\_\_\_\_

Note: no expected *value* here. But expected *utility* works fine.

# Why Expected Utility?

Do people really rank lotteries by expected utility?

## **Theorem (von Neumann & Morgenstern, 1944)**

If your ranking of lotteries satisfies a few “rationality” axioms, then there is a utility index  $u(o)$  such that

$$L \succ L' \iff U(L) > U(L') \quad \text{and} \quad L \sim L' \iff U(L) = U(L')$$

In words: you act *as if* you're maximizing expected utility

The axioms are in Bonanno §5.3. We're going to skip them.

- Interested? Read through them for fun!

From here on: we **assume** players maximize expected utility

# von Neumann & Morgenstern

- **von Neumann:** Budapest math prodigy. Princeton IAS (with Einstein)
  - Manhattan Project, modern computer design, minimax theorem
- **Morgenstern:** Vienna economist. Left Austria in 1938
- **1944:** *Theory of Games and Economic Behavior*. Game theory is born
- Expected utility axioms: an appendix to the 1947 edition!

## **Rescaling the Utility Index**

# Can We Rescale Any Way We Want?

**Chapter 2:** utility only *ranked* outcomes. Any increasing rescaling was fine.

Is that still true? Try squaring Alex's utility index:  $v(o) = u(o)^2$

|                 |           |             |             |
|-----------------|-----------|-------------|-------------|
| Outcome $o$     | Cazuela's | Yoshi Sushi | Dining hall |
| $u(o)$          | 10        | 6           | 0           |
| $v(o) = u(o)^2$ | 100       | 36          | 0           |

Using  $v$ :  $V(L_A) =$

$V(L_B) =$

# Can We Rescale Any Way We Want?

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Using  $v$ :  $V(L_A) =$   $V(L_B) =$

Same ranking of *outcomes*. Different ranking of *lotteries*!!

**Lesson:** The utility index can **not** be rescaled arbitrarily

# Linear Rescaling

Which rescalings *are* allowed? Only **linear** ones:

$$v(o) = a \cdot u(o) + b \quad (\text{with } a > 0)$$

- You can **multiply** by a positive number  $a$
- You can **add** (or subtract) a constant  $b$

Try  $v(o) = 2u(o) + 3$  on Alex's index (10, 6, 0):

$$v = ( \quad ) \quad V(L_A) = \quad \quad V(L_B) =$$

**Theorem:**  $u$  and  $v$  rank all lotteries the same way  $\iff v$  is a linear rescaling of  $u$

Like temperature:  $^{\circ}\text{F} = 1.8 \times ^{\circ}\text{C} + 32$ . Different numbers, same information.

Bonanno calls a linear rescaling a “positive affine transformation.”

# Normalizing

Since linear rescaling is allowed, we can always pick a convenient scale:

$$u(\text{best outcome}) = 1$$

$$u(\text{worst outcome}) = 0$$

This is called the **normalized** utility index. How to get it:

1. Subtract  $u(\text{worst})$  from every number (now the worst is 0)
2. Divide every number by the new  $u(\text{best})$  (now the best is 1)

|              |       |       |       |       |
|--------------|-------|-------|-------|-------|
| Outcome      | $o_1$ | $o_2$ | $o_3$ | $o_4$ |
| $u(o)$       | 8     | 2     | -2    | 0     |
| After step 1 |       |       |       |       |
| After step 2 |       |       |       |       |

# What the Normalized Numbers Mean

With a normalized index, the numbers have a nice interpretation.

Take a lottery over just the best and worst outcomes:

$$L = \begin{pmatrix} o_{\text{best}} & o_{\text{worst}} \\ p & 1 - p \end{pmatrix} \Rightarrow U(L) = p(1) + (1 - p)(0) = p$$

So  $u(o) = 0.4$  means:

- Getting  $o$  for sure is exactly as good as a **40% chance at the best outcome** (and a 60% chance of the worst)

This also tells us how to *find* someone's utility index. Just ask them:

“What chance  $p$  of the best outcome is just as good as  $o$  for sure?”

Their answer is  $u(o)$ .

# **Attitudes Toward Risk**

# Risk Averse, Risk Neutral, Risk Loving

Back to **money** lotteries, where  $E[L]$  exists. Offer someone a choice:

the lottery  $L$       vs.      getting  $\$E[L]$  for sure

## Definition

- **Risk averse:** prefers  $\$E[L]$  for sure
- **Risk neutral:** indifferent between them
- **Risk loving:** prefers the lottery  $L$

**Ex:**  $L_1 = \begin{pmatrix} \$10 & \$20 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$  vs.  $\$15$  for sure. Which would you take?

**Important:** This only makes sense when outcomes are **numbers** (like money).  
No  $E[L]$  means no “risk averse” or “risk loving.” (Bonanno doesn’t say this.)

# Risk Attitudes and the Shape of $u(o)$

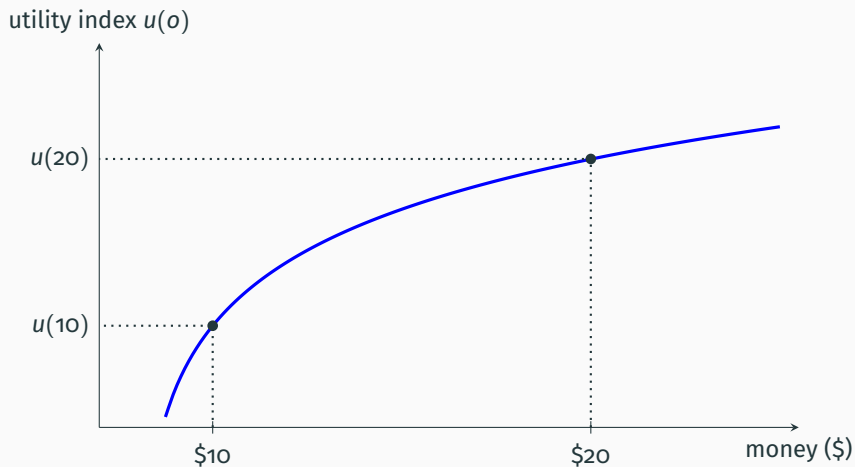
With expected utility, the definition becomes a comparison of two numbers:

$$\underbrace{U(L)}_{\text{expected utility of the lottery}} \quad \text{vs.} \quad \underbrace{u(E[L])}_{\text{utility of the expected value}}$$

|                     |        |                  |        |                          |
|---------------------|--------|------------------|--------|--------------------------|
| <b>Risk averse</b>  | $\iff$ | $U(L) < u(E[L])$ | $\iff$ | $u(o)$ is <b>concave</b> |
| <b>Risk neutral</b> | $\iff$ | $U(L) = u(E[L])$ | $\iff$ | $u(o)$ is <b>linear</b>  |
| <b>Risk loving</b>  | $\iff$ | $U(L) > u(E[L])$ | $\iff$ | $u(o)$ is <b>convex</b>  |

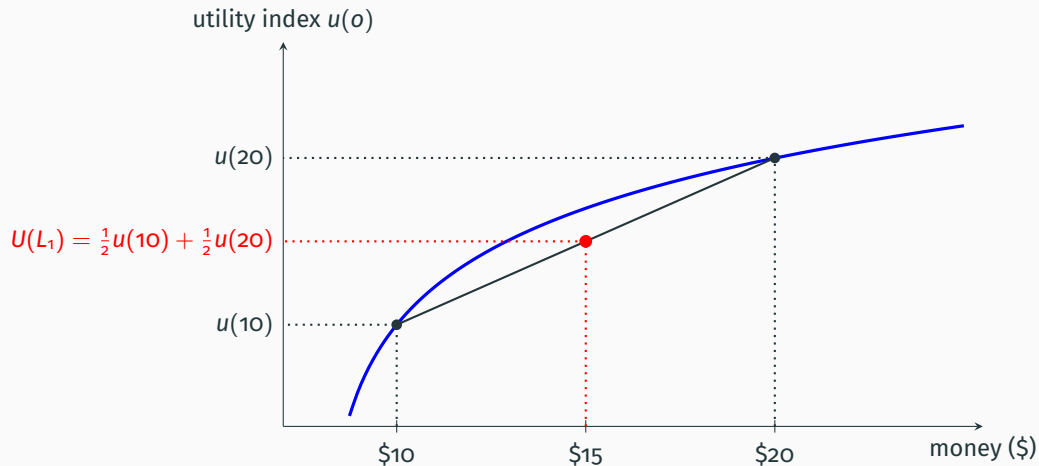
This isn't in Bonanno, but it's how economists usually think about risk attitudes.

# Risk Averse $\iff$ Concave



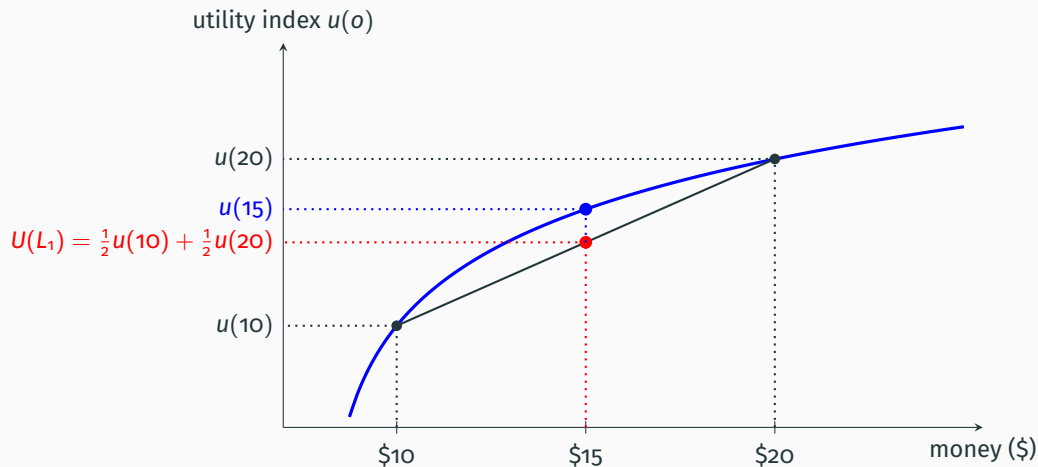
$L_1$  pays \$10 or \$20 with equal chance.  $E[L_1] = \$15$ .

## Risk Averse $\iff$ Concave



$U(L_1)$  is halfway up between  $u(10)$  and  $u(20)$ : the midpoint of the straight line.

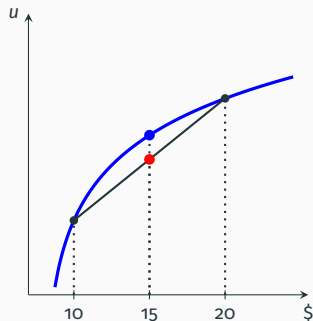
# Risk Averse $\iff$ Concave



The curve is *above* the line, so  $u(15) > U(L_1)$ . They'd rather have \$15 for sure.

# All Three Cases

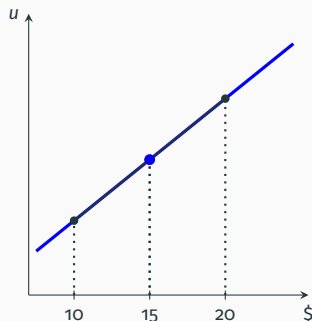
## Concave



$$U(L_1) < u(15)$$

Risk averse

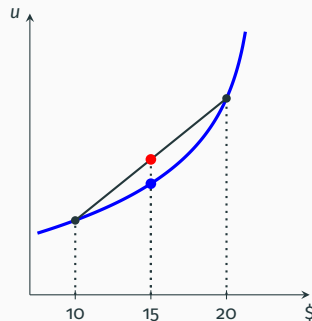
## Linear



$$U(L_1) = u(15)$$

Risk neutral

## Convex



$$U(L_1) > u(15)$$

Risk loving

**Red dot:** expected utility of the lottery. **Blue dot:** utility of \$15 for sure.

## Back to the Bomb Game

Open  $k$  boxes: you earn  $k$  with probability  $\frac{100-k}{100}$ , otherwise 0

Expected value:  $E[L_k] = k \cdot \frac{100 - k}{100}$

- Biggest when  $k =$  \_\_\_\_\_
- Opening **exactly** that many: risk \_\_\_\_\_
- Opening **fewer**: risk \_\_\_\_\_
- Opening **more**: risk \_\_\_\_\_

Our class median: \_\_\_\_\_

**Lab result:** the typical person opens a bit fewer than 50

Source: Crosetto & Filippin (2013), *Journal of Risk and Uncertainty*.

# Risk Attitudes in the Real World

Lots of real decisions boil down to “lottery vs. sure thing”:

- **Insurance.** Pay a premium for sure, or risk a big loss?
- **Extended warranties.** Same thing, for your laptop
- **Salary vs. commission.** Steady paycheck, or pay that depends on sales?
- **Index funds vs. single stocks.** Spread the risk, or bet on one company?
- **Sports betting, casinos, Ohio Lottery.** Pay to *take on* risk!

Real versions are messier:

- The same person might buy insurance *and* lottery tickets
- People treat gains and losses differently

But the core idea is the same: **the shape of  $u(o)$  tells you how they handle risk**

## **Example: Car Insurance**

# Should You Insure Your Car?

Ruth and Logan are OSU students. Each of them has:

- \$1,000 in the bank, and a car worth \$15,000 (total wealth: \$16,000)
- A 25% chance of totaling the car this year

Nationwide offers **full insurance** for a \$6,000 premium:

- If the car is totaled, they pay you the full \$15,000

Final wealth (in \$1,000s):

|              | Car is fine (75%) | Car is totaled (25%) |
|--------------|-------------------|----------------------|
| No insurance | 16                | 1                    |
| Insurance    | 10                | 10                   |

$E[\text{No insurance}] =$  \_\_\_\_\_ A risk-neutral person would choose: \_\_\_\_\_

## Two Different Utility Indices

Let  $o$  be final wealth (in \$1,000s).

**Ruth:**  $u_R(o) = \sqrt{o}$

$$U_R(\text{No ins.}) = \frac{3}{4}\sqrt{16} + \frac{1}{4}\sqrt{1}$$

$$u_R(10) = \sqrt{10} \approx 3.16$$

Ruth chooses: \_\_\_\_\_

Handy:  $\ln(16) \approx 2.77$  and  $\ln(1) = 0$ .

**Logan:**  $u_L(o) = \ln(o)$

$$U_L(\text{No ins.}) = \frac{3}{4}\ln(16) + \frac{1}{4}\ln(1)$$

$$u_L(10) = \ln(10) \approx 2.30$$

Logan chooses: \_\_\_\_\_

## Two Different Utility Indices

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**Logan:**  $u_L(o) = \ln(o)$

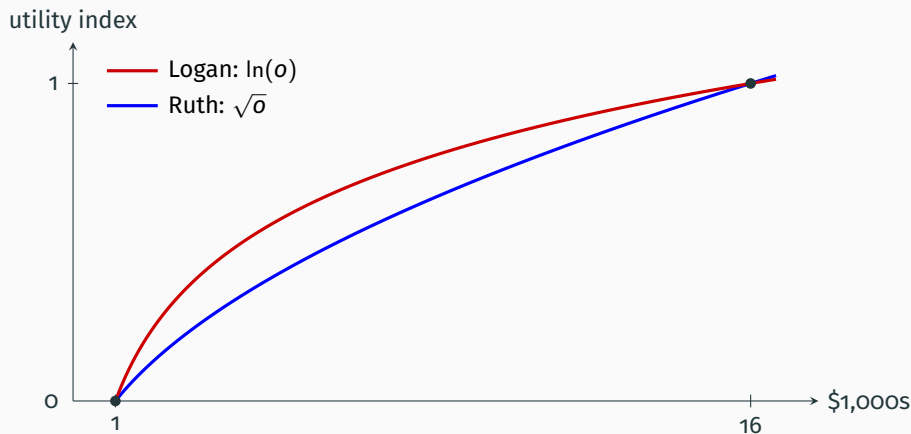
$$U_L(\text{No ins.}) = \frac{3}{4}\ln(16) + \frac{1}{4}\ln(1)$$

$$u_L(10) = \ln(10) \approx 2.30$$

Logan chooses: \_\_\_\_\_

Both are concave, so **both** are risk averse. But only one buys insurance!

# Seeing It in a Graph

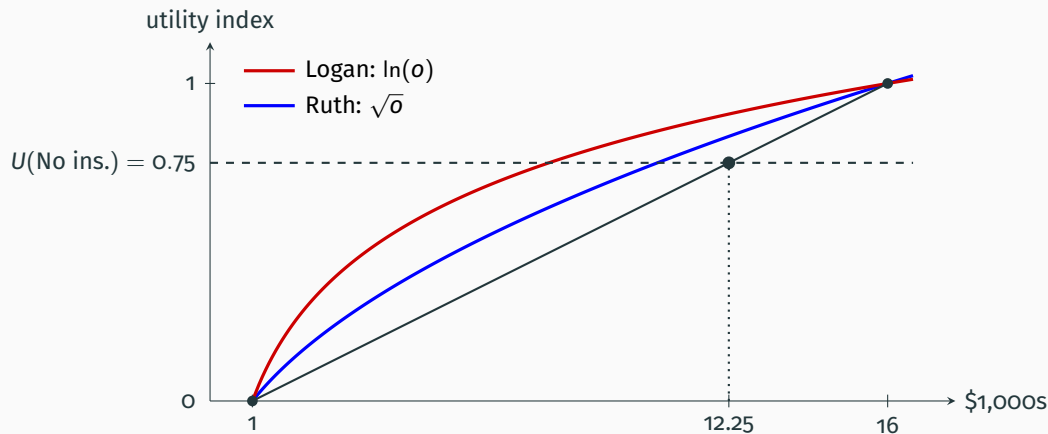


Both rescaled so that  $u(1) = 0$  and  $u(16) = 1$ . (Linear rescaling: allowed!)

Logan's curve bends more: Logan is **more** risk averse than Ruth.

Normalized: Ruth  $u_R(o) = \frac{1}{5}\sqrt{o} - \frac{1}{5}$  Logan  $u_L(o) = \frac{1}{5.36}\ln(o) \approx 0.36\ln(o)$

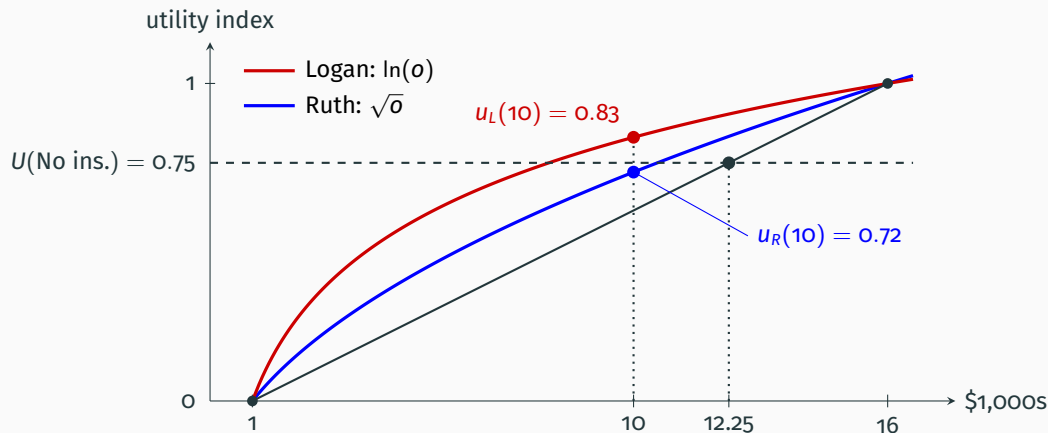
# Seeing It in a Graph



No insurance: 75% chance of the best, 25% chance of the worst.  
So  $U(\text{No ins.}) = 0.75$  for **both** of them: the dashed line.

Normalized: Ruth  $u_R(o) = \frac{1}{2}\sqrt{o} - \frac{1}{2}$  Logan  $u_L(o) = \frac{1}{\ln(16)} \ln(o) \approx 0.36 \ln(o)$

# Seeing It in a Graph



Insurance gives 10 for sure. Is  $u(10)$  above or below the dashed line?

**Ruth:** below, so no insurance. **Logan:** above, so buy insurance.

Normalized: Ruth  $u_R(o) = \frac{1}{2}\sqrt{o} - \frac{1}{2}$  Logan  $u_L(o) = \frac{1}{\ln(16)} \ln(o) \approx 0.36 \ln(o)$

## How Much Would They Pay?

The expected loss is  $\frac{1}{4} \times \$15,000 = \$3,750$ . Nationwide charges \$6,000.

What's the **most** each would pay for full insurance?

- **Ruth:** find the sure wealth  $w$  with  $\sqrt{w} = 3.25$
- **Logan:** find the sure wealth  $w$  with  $\ln(w) = \frac{3}{4} \ln(16)$

On the graph: where the dashed line crosses each curve.

**Lesson:** Knowing someone is risk averse doesn't tell you what they'll choose. You need to know their utility index.

**Let's Work One Out**

# The Prize Wheel

Carmen gets one spin of a prize wheel at the Ohio Union. The possible prizes:

- $M$ : a ticket to the Michigan game       $J$ : a pint of Jeni's ice cream
- $D$ : a dozen Buckeye Donuts       $N$ : nothing

Carmen's ranking:  $M \succ J \succ D \succ N$ . Carmen also tells us:

- $J$  for sure is just as good as a 50% chance of  $M$  (otherwise  $N$ )
- $D$  for sure is just as good as a 20% chance of  $M$  (otherwise  $N$ )

Carmen can pick which wheel to spin:

$$W_1 = \begin{pmatrix} J & D \\ 0.5 & 0.5 \end{pmatrix} \quad W_2 = \begin{pmatrix} M & N \\ 0.3 & 0.7 \end{pmatrix} \quad W_3 = \begin{pmatrix} M & J & D \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

# The Prize Wheel

1. Write down Carmen's normalized utility index.
2. Find the expected utility of each wheel. Which does Carmen spin?
3. Rescale using  $v(o) = 10u(o) + 5$ . Does the ranking of wheels change?
4. Now rescale using  $w(o) = u(o)^2$ . Does the ranking change? Why?
5. Is Carmen risk averse, risk neutral, or risk loving?

# What Have We Learned, Charlie Brown?

- **Expected value** needs numeric outcomes. **Expected utility** doesn't
- $u(o)$ : **utility index** over outcomes
- $U(L)$ : **expected utility** over lotteries
  - $U(L) = p_1 u(o_1) + \dots + p_n u(o_n)$
- The utility index can only be rescaled **linearly**
  - So we can normalize: best outcome = 1, worst outcome = 0
- Risk averse / neutral / loving  $\iff u(o)$  is concave / linear / convex
  - Only for numeric outcomes!
- Two risk-averse people can still make different choices

Next: games where the payoffs are expected utilities

- Bonanno, *Game Theory*, Chapter 5
  - §5.1 (risk attitudes) and §5.2 (expected utility)
  - §5.3 (the axioms): optional, read it for fun
- Remember: Bonanno's  $U(o)$  is our  $u(o)$ . His  $E[U(L)]$  is our  $U(L)$
- Exercises: §5.4.1 (risk attitudes) and §5.4.2 (expected utility)
  - Exercises 5.2 and 5.4: outcomes that aren't numbers
  - Exercise 5.11: two risk-averse people who choose differently
- Solutions in §5.5